3.2 Pathogens, immunity and disease control

In this section, learners study the nature of pathogens and their means of transmission. An understanding of the primary defences and immunological responses of the body to infection, and the management and treatment of communicable diseases will also be considered, including the challenges of controlling the spread of drug-resistant strains. Learners will also gain an appreciation of the need for epidemiological research in this dynamic and changing area of science, and evaluate the associated risks and benefits to the individual and to society.

Learners are expected to apply knowledge, understanding and other skills gained in this section to new situations and/or to solve related problems.

3.2.1 Pathogenic microorganisms

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	how pathogens (including bacteria, viruses and fungi) cause communicable disease	To include an outline of the general mechanisms of pathogenicity by bacteria (toxin production), viruses (taking over cell metabolism) and fungi (enzyme secretion).
(b)	the causes, means of transmission, symptoms and the principal treatment of tuberculosis (TB) and HIV/AIDS	To include droplet infection, details of primary and secondary TB and also opportunistic infections (HIV-AIDS). HSW10

(c)	the structure of the Human Immunodeficiency Virus (HIV)	To include the use of diagrams showing the location of enzymes and the nature of the genetic material.
(d)	**(i)** the use of Gram stain, cell and colony morphology to identify bacteria	
	(ii) the culturing of bacteria and the identification of Gram-positive and Gram-negative bacteria using the Gram staining method on pure cultures	HSW3, HSW4, HSW5, HSW6, HSW8
(e)	how the incidence and prevalence of a communicable disease can change over time	To include the principles of endemic communicable diseases (e.g. chickenpox in the UK), epidemics (e.g. SARS in China, 2002) and pandemics (e.g. H1N1 influenza in 2009).
(f)	calculations of incidence rates, prevalence rates and mortality rates and their importance in epidemiology	*M0.1, M0.3, M0.4*
(g)	the analysis, interpretation and use of epidemiological data	To include the evaluation of graphical data to assess the impact of disease e.g. for HIV and TB infection. *M1.3, M1.7, M3.1, M3.5, M3.6* HSW12
(h)	the importance of reporting notifiable diseases and the role of Public Health England, formerly known as the Health Protection Agency (HPA)	To include examples of notifiable diseases. HSW12
(i)	the social, ethical, economic and biological factors involved in the attempts to control and prevent diseases in the context of HIV/AIDS and TB.	HSW10